

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence
and Emerging Technologies

DELIVERABLE No. 4 – MINI SYSTEMATIC REVIEW, PRISMA 2020 & GAP ANALYSIS

Research topic: Artificial intelligence governance maturity model for Peruvian organizations within the framework of Law No. 31814 and NTP-ISO/IEC 42001:2025: design, validation, and automated compliance assessment.

Review question: What organizational AI/governance dimensions, maturity levels, assessment mechanisms, and implementation practices are reported in the literature, and what gaps remain for developing and validating a context-specific maturity model for Peruvian organizations?

ARTIFACT 1. MINI SYSTEMATIC REVIEW

Documented search strategy. Target database: Semantic Scholar. Proposed search date: 2 October 2026. Search string: (“AI governance” OR “responsible AI governance” OR “artificial intelligence governance”) AND (“maturity model” OR maturity synonyms to be grouped with OR and concept blocks joined with AND. For the mini-review, the first 30 relevant records should be screened by title/abstract and the final 10 selected using the criteria below.

Inclusion criteria: English or Spanish; scholarly article/review; organizational or public-sector AI governance; maturity, assessment, implementation, controls, or governance framework; relevance to model dimensions, maturity levels, indicators, validation, regulatory alignment, or operationalization.

Exclusion criteria: Purely technical AI/model-performance studies; governance only at geopolitical level without organizational implications; non-AI governance; duplicates; sources without sufficient methodological or conceptual detail.

# / Study	Main contribution	Relevance to thesis	Decision
1. Mäntymäki, M., Minkinen, M., Birkstedt, T., & Viljanen, M. (2022). Defining organizational AI governance. <i>AI and Ethics</i> , 2, 603–609. https://doi.org/10.1007/s43681-022-00143-x	Defines organizational AI governance as an integrated system of rules, practices, processes and technological tools aligned with strategy, law and ethical principles.	Conceptual foundation for governance dimensions and organizational scope.	Include
2. Papagiannidis, E., Enholm, I. M., Dremel, C., Mikalef, P., & Krogstie, J. (2023). Toward AI Governance: Identifying Best Practices and Potential Barriers and Outcomes. <i>Information Systems Frontiers</i> , 25, 123–141. https://doi.org/10.1007/s10796-022-10251-y	Comparative organizational evidence identifies governance practices, barriers and outcomes associated with robust AI applications.	Supports measurable organizational practices and implementation indicators.	Include
3. Birkstedt, T., Minkinen, M., Tandon, A., & Mäntymäki, M. (2023). AI governance: themes, knowledge gaps and future agendas. <i>Internet Research</i> , 33(7), 133–167. https://doi.org/10.1108/INTR-01-2022-0042	PROSMA-informed review identifies technology, stakeholders/context, regulation and processes, plus major empirical and operationalization gaps.	Direct evidence for knowledge, contextual and methodological gaps.	Include
4. Mylrea, M., & Robinson, N. (2023). Artificial Intelligence (AI) Trust Framework and Maturity Model: Applying an Entropy Lens to Improve Security, Privacy, and Ethical AI. <i>Entropy</i> , 25(10), 1429. https://doi.org/10.3390/e25101429	Proposes an AI trust framework and maturity model integrating security, privacy and ethical AI considerations.	Provides maturity-model structure and assessment concepts.	Include
5. Batool, A., Zowghi, D., & Bano, M. (2025). AI governance: a systematic literature review. <i>AI and Ethics</i> , 5, 3265–3279. https://doi.org/10.1007/s43681-024-00653-w	Systematic review of 28 studies organizes AI governance around what, when and how, across team, organization, industry, national and international levels.	Useful taxonomy for model dimensions and governance artifacts.	Include
6. Papagiannidis, E., et al. (2025). Responsible artificial intelligence governance: A review and research framework. <i>Journal of Strategic Information Systems</i> , 34(2), 101885. https://doi.org/10.1016/j.jsis.2024.101885	Review develops a responsible-AI governance framework centered on structural, procedural and relational practices and notes limited operationalization	Supports governance dimensions and highlights principles-to-practice gap.	Include
7. Almeida, P. G. R. de, & dos Santos Júnior, C. D. (2024). Artificial intelligence governance: Understanding how public organizations implement it. <i>Government Information Quarterly</i> , 41, 102003. https://doi.org/10.1016/j.giq.2024.102003	Empirical study of 28 public organizations identifies combinations of processes and practices and proposes the AIGov4Gov framework.	Relevant to public organizations and empirical implementation evidence.	Include
8. Ferreira, R. M. F. D., Grilo, A., & Maia, M. (2025). Piloting a maturity model for responsible artificial intelligence: A Portuguese case study. <i>Journal of Responsible Technology</i> , 22, 100117. https://doi.org/10.1016/j.jrt.2025.100117	Pilots an industry-wide Responsible AI maturity model in three companies and two research centers, demonstrating practical gap identification.	Strong methodological precedent for piloting and validating a maturity instrument.	Include

9. Heger, A. K., Passi, S., Dhanorkar, S., Kahn, Z., Wang, R., & Vorvoreanu, M. (2025). Towards a Responsible AI Organizational Maturity Model. Proceedings of the ACM on Human-Computer Interaction, 9(7), Article CSCW333. https://doi.org/10.1145/3757514	Developed from interviews and co-design with 90 Responsible-AI experts; proposes 24 dimensions across organizational foundations, team approach and RAI practices.	Direct benchmark for dimensions, maturity levels and expert-based model development.	Include
10. Huancapaza Hilasaca, J. E. (2025). Implementación de inteligencia artificial en el Estado peruano: catálogo analítico de aplicaciones (2025). Interfases, 22. https://doi.org/10.26439/interfases2025.n022.8263	Mpas 22 AI applications in 17 Peruvian public entities and reports limited documentation on transparency, bias auditing and rights-impact assessment.	Provides Peru-specific contextual evidence supporting the need for governance maturity assessment.	Include

Mini-review synthesis. The selected literature converges on AI governance as an organizational capability that must translate high-level ethical and regulatory principles into repeatable structures, processes, roles, controls, evidence and lifecycle practices. Recent maturity-model studies provide multidimensional structures and tagged progression, while empirical work shows that implementation remains uneven and context-dependent. The literature therefore supports constructing a maturity model that combines organizational foundations, operational governance practices, accountability, risk management, transparency, human oversight, monitoring and continuous improvement. However, the evidence base remains concentrated outside Peru, and no included study validates a maturity model specifically against the Peruvian regulatory framework and NTP-ISO/IEC 42001:2025 or evaluates LLM-assisted documentary compliance assessment as part of the same maturity architecture.

ARTIFACT 2. PRISMA 2020 FLOW – MINI REVIEW

Important validation note: The numbers below form a consistent mini-review workflow based on the Session 4 instruction to screen 30 results and retain 10.

PRISMA phase	Proposed count	Documented action / reason
Identification	n=30 records exported for mini-review screening	First 30 potentially relevant Semantic Scholar records from the documented query; duplicates checked before screening.
Screening	n=30 titles/abstracts screened; n=16 excluded	Exclude purely technical AI studies, non-organizational governance, non-AI governance, and clearly irrelevant contexts.
Eligibility	n=14 reports assessed for eligibility n=4 excluded	Exclude insufficient conceptual/methodological detail (2), weak fit with organizational governance/maturity (1), and inaccessible/insufficient evidence (1).
Included	n=10 studies included	Final studies listed in Artifact 1.

ARTIFACT 3. GAP ANALYSIS TABLE

Gap type	Candidate gap for the thesis	Evidence from reviewed literature	How the doctorate addresses it
Knowledge	Limited empirical knowledge of how organizational AI-governance maturity can be measured as a distinct, multidimensional capability and linked to evidence of effective governance.	Birkstedt et al. (2023) report limited empirical understanding of AI-governance implementation and insufficient operationalization; Papagiannidis et al. (2025) describe a principles-to-practice gap.	Develop and validate measurable dimensions.
Methodological	Few maturity models combine expert-derived dimensions, psychometric validation, multi-organization application and automated documentary assessment in one evaluated design.	Ferreira et al. (2025) pilot a practical maturity model in five Portuguese organizations; Heger et al. (2025) derive 24 dimensions through qualitative expert work, indicating complementary but different validation approaches.	Use DSR + Delphi + Aiken's V + reliability analysis + organizational survey + human-versus-LLM agreement evaluation.
Contextual	Organizational AI-governance maturity has not been validated for Peruvian public and private organizations under Peru's specific regulatory and standards environment.	Huancapaza Hilasaca (2025) documents 22 AI applications in 17 Peruvian public entities and identifies governance-documentation weaknesses; most organizational maturity evidence in the selected corpus comes from other countries.	Contextualize the model to Law No. 31814 and NTP-ISO/IEC 42001:2025 and establish a Peruvian multi-organization baseline.

Theoretical	Existing governance frameworks explain principles, practices or maturity dimensions, but there is no integrated explanatory architecture linking regulatory alignment, organizational capability maturity, documentary evidence and automated assessment.	Mäntymäki et al. (2022) position AI governance within organizational governance; Batool et al. (2025) provide multi-level governance taxonomy; maturity studies add staged progression but remain fragmented.	Integrate governance dimensions, maturity progression, regulatory/standard requirements and evidence-based assessment into a single artifact-centered model.
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Provisional conclusion: The strongest doctoral opportunity is the intersection of the contextual and methodological gaps: adapting the emerging organizational AI-governance maturity literature to Peru and validating it rigorously, while adding an LLM-assisted evidence-assessment artifact. This keeps the research aligned with the Design Science Research paradigm defended in Deliverables 1 and 2 and with the PRISMA-based literature -review stage specified in Deliverable 3.

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